Stress-testing Deep Learning Models on Sparse Laboratory Data

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***Abstract*—Temporal deep learning models have demonstrated strong performance on clinical time-series data in ICU settings, where measurements are dense and frequently recorded. However, real-world non-ICU data are often sparse, irregular, and highly imbalanced, raising fundamental questions about model robustness under these conditions. In this study, we systematically evaluate the robustness of five temporal deep learning architectures (GRU, GRU-D, TCN, SaND, and STraTS) under three controlled perturbation regimes that mimic non-ICU characteristics: reduced cohort size, increased observation sparsity, and severe class imbalance. Using the standard ICU benchmarks MIMIC-III and PhysioNet Challenge 2012, all models are trained under a consistent preprocessing and training protocol. Our results reveal that dataset size and class imbalance are the dominant drivers of performance degradation. Performance on PhysioNet-2012 drops substantially below approximately 2,000 training samples, while AUROC declines sharply when the positive class falls below 20% of the training set. Observation sparsity has a comparatively smaller effect on median performance, but introduces notable instability across runs. Notably, architectures specifically designed for irregular time-series data, such as GRU-D and STraTS, do not consistently outperform simpler recurrent or convolutional baselines under these extreme conditions, and GRU-D exhibits a distinct failure mode under severe sparsification due to its reliance on temporal gap computations. These findings suggest that data characteristics, rather than model architecture complexity, are the primary determinants of clinical prediction performance in real-world sparse settings, with important implications for the design and deployment of machine learning systems in non-ICU healthcare environments.**

***Index Terms*—Temporal deep learning, clinical time-series, sparse data, class imbalance, electronic health records, model robustness, stress testing**

I. Introduction

Laboratory measurements are numerical observations obtained from clinical tests (e.g., blood or urine) and are traditionally used for diagnosis and monitoring. More recently, they have also been used as input features for machine learning models to perform predictive tasks such as mortality prediction and disease progression modeling [1]. When collected over time, these measurements form multivariate clinical time-series data, where each patient is represented by a sequence of temporally ordered observations [2]. A key characteristic of clinical time-series data is irregular sampling. Measurements are recorded based on clinical decisions, leading to heterogeneous observation frequencies across variables [3]. Furthermore, missing values are often informative rather than random, reflecting that a test was not considered necessary at a given time [4]. These properties differ significantly across healthcare settings. ICU data are collected in high-acuity environments with frequent monitoring, resulting in relatively dense and information-rich time series [5]. In contrast, non-ICU data are sparser, more irregular, and typically smaller in size, with fewer observations per patient [6]. Additionally, outcomes in non-ICU settings are often highly imbalanced, making prediction tasks more challenging [7]. Such data characteristics directly impact model performance. Temporal deep learning models often rely on sufficient data density and balanced distributions, and may fail to generalize under sparse and imbalanced conditions, where simpler methods can sometimes perform better [8].

ICU clinical time-series data are characterized by frequent monitoring, heterogeneous measurement rates, irregular sampling, and informative missingness [9]. These data are highly complex, with fine temporal granularity and multimodal information. For example, MIMIC-III includes diverse data sources such as vital signs, laboratory results, medications, procedures, and clinical notes [10]. Due to rapidly changing patient conditions, measurements are recorded repeatedly, resulting in dense longitudinal records with varying levels of detail across patients and variables. Additionally, ICU data exhibit substantial heterogeneity across hospitals and care settings, as shown by the eICU database [11]. Overall, ICU time-series data are temporally rich, irregularly sampled, and diverse in both content and structure.

Non-ICU clinical data are heterogeneous and challenging to use, as they are primarily collected for routine care rather than research. Holmes et al. [12] note that electronic health records are shaped by clinical workflows, billing processes, and institutional practices, resulting in variation in data structure, completeness, and consistency across patients and settings. As a result, the same clinical concept may be recorded differently across institutions. Additionally, the presence and timing of observations in non-ICU data are inherently informative. Sisk et al. [13] highlight that measurements are collected irregularly, depending on patient interactions and clinical decisions. Consequently, observation patterns reflect factors such as disease severity, making non-ICU data irregular, unevenly sampled, and influenced by the care process.

Most temporal deep learning research in healthcare has focused on ICU benchmarks, where data are dense and frequently recorded. Prior work suggests that these models may not generalize well to non-ICU cohorts, which are typically

sparser, more irregular, and more imbalanced.

Motivated by this gap, we investigate whether data characteristics themselves drive performance degradation. To this end, we systematically stress-test temporal models under controlled perturbations that mimic non-ICU conditions, including reduced cohort size, increased sparsity, and class imbalance.

Unlike prior work that primarily reports performance differences between ICU and non-ICU settings, we isolate the impact of individual data characteristics—dataset size, sparsity, and class imbalance—through controlled perturbations of standard benchmark datasets. This enables direct attribution of performance degradation to specific factors, rather than confounding effects across datasets.

This motivates the following research questions:

RQ1 **(Sparsity)** How does model performance change as benchmark trajectories are progressively sparsified?

RQ2 **(Dataset size)** How sensitive are temporal models to reductions in training data?

RQ3 **(Class imbalance)** How does severe class imbalance affect performance and stability across runs?

Objectives: To answer these questions, we stress-test temporal deep learning models by progressively modifying benchmark cohorts along three axes:

- **Dataset size:** subsampling to smaller cohorts.
- **Sparsity:** structured sparsification by jointly masking TS-ID and variable ID (TS-ID + Var-ID).
- **Class imbalance:** constructing rare-positive settings.

Our goal is to use these controlled perturbations to identify thresholds in sparsity, sample size, and class imbalance beyond which deep learning models lose their advantage over simpler approaches, and to translate these findings into practical guidance for applying temporal deep learning to sparse longitudinal health data.

This study assumes that ICU benchmarks represent the high-data-density end of the clinical spectrum, and that controlled degradation of these datasets serves as a valid proxy for non-ICU conditions. The joint TS-ID and variable-ID sparsification strategy is assumed to approximate the irregular measurement patterns of outpatient settings, and in-hospital mortality and length-of-stay are treated as representative prediction tasks. These assumptions are motivated by structural similarities between perturbed ICU data and real non-ICU cohorts such as neutropenic fever patients.

II. Method

A. *Datasets and Tasks*

We follow the standard pipeline proposed by Tipirneni et al. [1] and use ICU EHR benchmarks, where each sample corresponds to a single ICU stay with multivariate clinical measurements recorded at irregular time intervals, together with demographic variables. We consider two datasets, MIMIC-III and PhysioNet Challenge 2012, and adopt the same cohort construction, feature extraction, and preprocessing conventions as in the Tipirneni et al. pipeline [1].

MIMIC-III: MIMIC-III is a large, publicly available critical care database containing adult ICU stays. Following the preprocessing pipeline of Tipirneni et al. [1], we extract a set of 129 clinical variables from multiple event sources, including laboratory measurements, charted vital signs, medications/prescriptions, and input/output events. Measurements are retained with their original timestamps, resulting in irregularly sampled trajectories with substantial missingness across variables and time. Demographic and static covariates are included alongside the time-varying features as in the same pipeline.

PhysioNet Challenge 2012. The PhysioNet 2012 ICU dataset provides multivariate time series for adult ICU stays including demographic information. In the configuration of Tipirneni et al. [1], each stay contains 37 time-series variables, and observations are recorded sparsely and at variable-specific sampling frequencies. As with MIMIC-III, trajectories are treated as irregular event streams rather than fully observed grids, and the dataset exhibits high rates of missing values that reflect real clinical measurement practices.

Prediction tasks: Following the task definitions used in the experimental setup of Tipirneni et al. [1], we consider two clinical prediction tasks on both datasets.

First, we evaluate **in-hospital mortality**, formulated as a binary classification problem using an early observation window (first 24 hours for MIMIC-III and first 48 hours for PhysioNet-2012) to predict whether the patient dies during the hospital stay.

Second, we evaluate the **length of stay (LOS)** prediction. Unlike the original formulation in Tipirneni et al. [1], where LOS is treated as a continuous variable, we convert LOS into a binary classification task. Specifically, a threshold T is computed from the training set to define a positive class corresponding to longer hospital stays, following the experimental setup described in our perturbation framework.

The same threshold is then applied unchanged to the validation and test sets to ensure consistency across splits. This formulation enables controlled analysis of model performance under varying levels of class imbalance while maintaining comparability across models and perturbation settings.

B. *Temporal Models*

We evaluate representative temporal deep learning architectures for clinical time-series prediction, covering recurrent, convolutional, and attention-based model families. All models are trained under the same preprocessing and training protocol (STraTS pipeline), while stress-testing is performed by perturbing only the input datasets [1].

GRU: The gated recurrent unit (GRU) is a recurrent neural network that models sequences through gated state updates, enabling the hidden state to retain or overwrite past information as needed [14]. We use GRU as a standard recurrent baseline.

GRU-D: GRU-D extends GRU to explicitly handle informative missingness and irregular sampling by incorporating

missingness masks, time-since-last-observation features, and learnable decay terms that modulate inputs and hidden states as time gaps increase [15].

TCN: As a convolutional baseline, we use a temporal convolutional network (TCN) based on stacked causal 1D convolutions (with dilation and residual connections), which enables long-range temporal modeling with parallel computation across time steps [16].

SaND: As an attention-based baseline, we include Simply Attend and Diagnose (SaND), which uses masked self-attention to capture global temporal dependencies for clinical time-series prediction [17].

STraTS: Finally, we include STraTS, which represents each patient trajectory as a set of observation triplets (t, v, x) and applies Transformer-style attention over triplet embeddings using continuous value embeddings for times and measurement values [1].

C. *Perturbation Strategies*

To evaluate model robustness under progressively harder data conditions, we create perturbed variants of each benchmark dataset while keeping the original data format and train/validation/test split structure unchanged. We apply three types of perturbations: *Subsampled* (reduced cohort size), *Sparsified* (increased measurement sparsity), and *Imbalanced* (increased label imbalance). Each variant is saved as a separate PKL file in the same tuple format as the original processed data, allowing it to be used as a drop-in replacement in the training pipeline [1].

1) ***Subsampled (reduced cohort size):*** In order to produce smaller reference groups, we downsample the training and validation admissions while keeping the test set fixed. For each subsampling level $p \in \{10, 20, \ldots, 90\}$ and random seed s, we randomly select $p\%$ of admission IDs from the original training split and $p\%$ from the original validation split (without replacement). We then restrict the time-series and outcome tables to the selected IDs together with all test IDs, and save the resulting dataset with updated split lists. This design isolates the effect of limited labeled data while ensuring that all models are evaluated on the same, unchanged test distribution.

2) ***Sparsified (increased observation sparsity):*** To mimic extremely sparse follow-up trajectories, we increase sparsity by removing time-series measurements while keeping demographic information and outcome labels unchanged. We consider a sparsification strategy controlled by a keep-percentage p and a random seed s. Measurements are sparsified in a systematic way by using the same method to retain data on each clinical variable during an admission. Only a fixed percentage of the measurements for each admission-variable pair is maintained, resulting in variable-specific sparsity levels. This indicates that some clinical factors are recorded more frequently than others. The same minimum-viability check is applied, and admissions that result in measurements at fewer than two time points are eliminated, with the split lists adjusted accordingly.

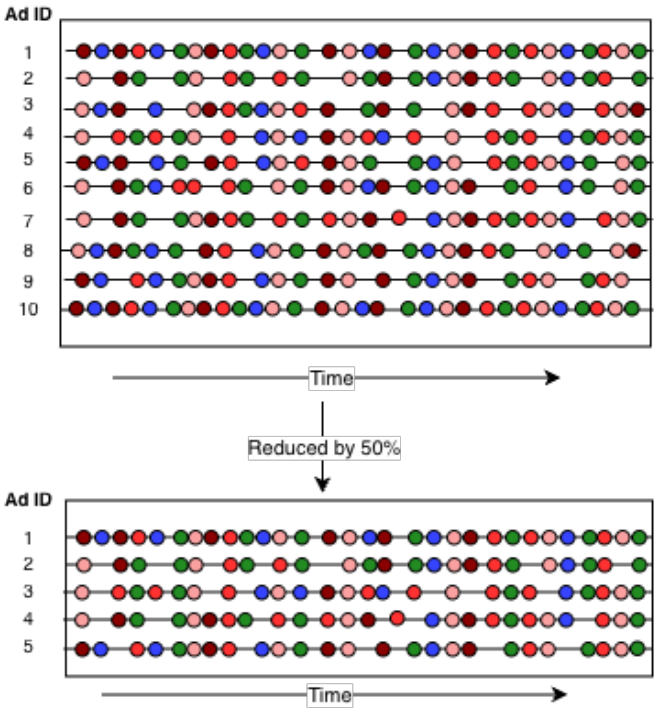

Fig. 1. Perturbation Approach (Subsampled)

3) ***Imbalanced (label imbalance via length-of-stay thresholding):*** In the length-of-stay (LOS) setting, label imbalance is created by converting LOS into a binary target using a threshold. A target positive-class rate p is chosen first, and the LOS threshold T is then computed from the training split only, so that approximately $p\%$ of training admissions are labeled as positive. After fixing T, the same threshold is applied unchanged to the validation and test splits, and the resulting class proportions are reported for each split. LOS for PhysioNet-2012 is taken directly from the results table, while LOS for MIMIC-III is calculated using the dataset's available admission time data before thresholding. To guarantee reproducibility, the chosen threshold T is saved.

The key methodological contribution is the controlled operationalization of non-ICU data challenges as reproducible perturbations of established ICU benchmarks. Unlike prior work that documents ICU-to-non-ICU performance gaps without isolating their causes, this framework applies each perturbation independently across a consistent severity range, enabling direct attribution of performance degradation to specific data characteristics.

III. Experimental Setup

To systematically evaluate the robustness of temporal deep learning models under non-ICU-like conditions, we construct controlled perturbations of the original ICU datasets. These perturbations simulate three key challenges observed in real-

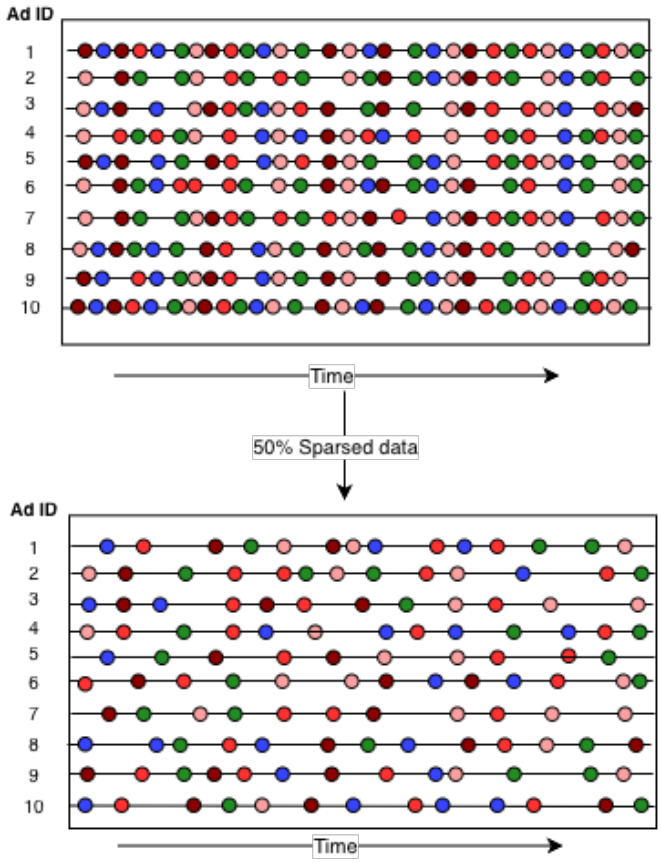

Fig. 2. Perturbation Approach (Sparsification)

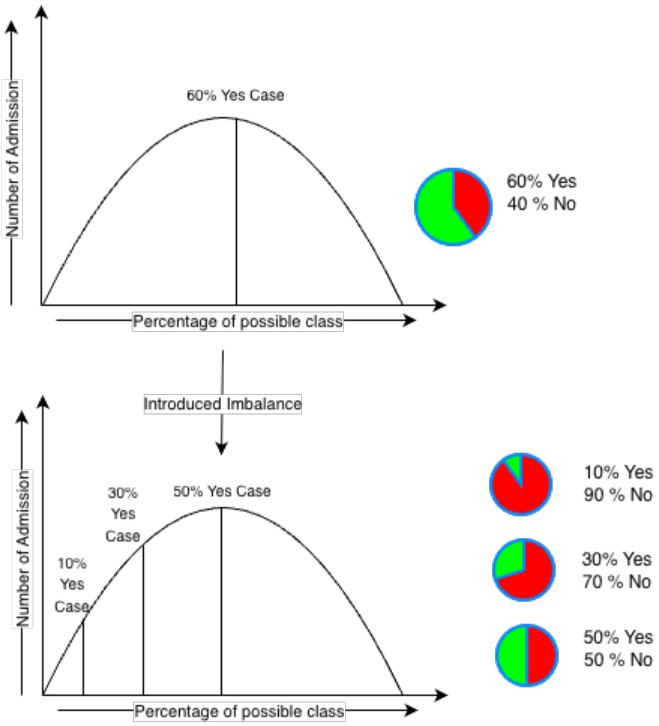

Fig. 3. Perturbation Approach (Imbalanced)

world clinical data: reduced cohort size, increased sparsity of measurements, and severe class imbalance. Importantly, all perturbed datasets preserve the original data format and train/validation/test split structure, ensuring fair and consistent evaluation across models.

*1) **Subsampled (Reduced Cohort Size)**:* To simulate smaller cohort sizes typical of non-ICU settings, we perform subsampling on the training and validation splits while keeping the test set fixed. For a given percentage $p \in \{1, 2, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$ and random seed s, we randomly select $p\%$ admission IDs from both the training and validation sets without replacement. The corresponding time-series and outcome data are then restricted to these selected admissions, while all test samples remain unchanged.

This setup isolates the effect of limited training data and allows us to analyze how model performance degrades as the available cohort size decreases.

*2) **Sparsification (Increased Observation Sparsity)**:* To mimic the sparse and irregular measurement patterns commonly found in non-ICU data, we introduce sparsification by removing a fraction of time-series observations. For each admission-variable pair, only $p\%$ of measurements is retained, where p follows the same range as in the subsampling experiment.

This approach preserves the structure of each clinical variable while reducing the number of observations per trajectory. Demographic features and outcome labels remain unchanged. A minimum viability constraint is enforced to ensure that each admission retains at least two time points; otherwise, the admission is removed from the dataset.

This perturbation allows us to evaluate the sensitivity of models to missing measurements and irregular sampling.

*3) **Imbalanced (Label Imbalance via Length-of-Stay Thresholding)**:* To study the impact of class imbalance, we construct binary classification targets from the continuous length-of-stay (LOS) variable. For a desired positive class rate $p \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30, 40, 50\}$, a threshold T is computed using only the training split such that approximately $p\%$ of admissions are labeled as positive.

Once the threshold T is determined, it is applied consistently to the validation and test splits without modification. This ensures that the evaluation reflects realistic distribution shifts encountered in practice.

This perturbation enables controlled analysis of model performance under increasingly imbalanced label distributions, which are common in non-ICU clinical applications.

*4) **Experimental Protocol**:* All models are trained using the same preprocessing pipeline and hyperparameter configurations to ensure comparability. For each perturbation setting, experiments are repeated across 10 random seeds (except for the imbalance setting, where thresholding is deterministic). We evaluate model robustness under three perturbation settings: subsampling, sparsification, and class imbalance. Table I summarizes the experimental setup.

TABLE I
EXPERIMENTAL SETUP FOR THE THREE PERTURBATION STRATEGIES.

Details	Subsampled	Sparsification	Imbalanced
Description	Reduces training and validation size by selecting a random subset of admission IDs.	Increases sparsity by removing time-series measurements while keeping demographics unchanged.	Creates class imbalance by converting continuous LOS into a binary target.
Target	In Hospital Mortality	In Hospital Mortality	Length of Stay
Percentage	1, 2, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100	1, 2, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100	1, 2, 3, 4, 5, 6, 7, 8, 9, 10,20, 30, 40, 50
Seed	10 random seeds	10 random seeds	None
Key Process	Select $p\%$ of admission IDs from training and validation; keep test fixed	Retain $p\%$ of measurements for each admission-variable pair.	Compute threshold T on training and apply it unchanged to validation and test.

IV. RESULTS

In this section, we present the results of the stress-testing experiments under three controlled perturbation settings: reduced cohort size (subsampling), increased sparsity (sparsification), and increased class imbalance (Imbalanced). All results are reported on the unchanged test set to ensure comparability across perturbation levels.

A. *Effect of Reduced Cohort Size (Subsampling)*

To evaluate the sensitivity of temporal models to limited training data, we progressively reduce the size of the training and validation cohorts while keeping the test set fixed. The percentage of retained data p is varied from 1% to 100%.

Across both datasets, model performance degrades consistently as training cohort size decreases, with a clear phase transition below 20% of the original data.

Physionet-2012: Figure 4 shows the impact of subsampling on Physionet-2012. All models achieve a mean AUROC above 0.82 with full data (100%), but decline to approximately 0.73–0.76 with 10% of the data (approximately 1,000 admissions) and below 0.70 at 1–2% (approximately 100–200 admissions). The sharpest drop occurs between 5% and 20%, corresponding to roughly 500–2,000 admissions.

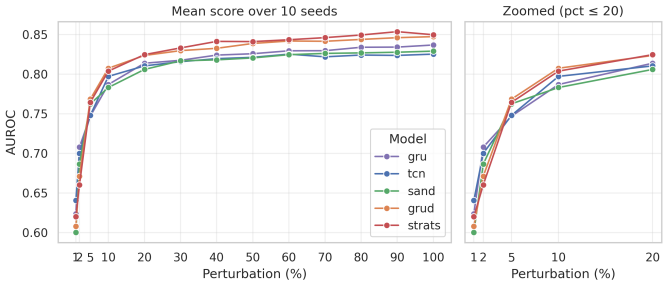

Fig. 4. Model performance (AUROC) under subsampling on Physionet-2012 across different cohort sizes.

MIMIC-III: As illustrated in Fig. 5, a similar pattern holds but the degradation threshold is higher, occurring between 10% and 20% of data (approximately 3,000–5,000 admissions), reflecting the greater complexity and variability of MIMIC-III trajectories compared to Physionet-2012.

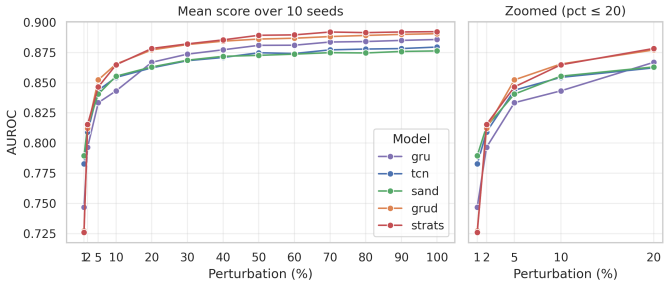

Fig. 5. Model performance (AUROC) under subsampling on MIMIC-III across different cohort sizes.

At extreme subsampling levels (1–5%), architectures specifically designed for irregular time-series (GRU-D, STraTS) degrade more rapidly than simpler baselines (GRU, TCN). For example, on Physionet-2012 at 2% data, STraTS AUROC drops to approximately 0.67 compared to 0.72 for standard GRU. This suggests that the additional inductive biases of irregular-time-series models (temporal decay, triplet attention) require sufficient data volume to provide benefit, and may become a liability in very low-data regimes.

These results show that temporal models are highly sensitive to cohort size, with reliable learning no longer achievable below approximately 2,000 samples on Physionet-2012 and 5,000 on MIMIC-III.

B. *Effect of Increased Sparsity (Sparsification)*

To simulate non-ICU conditions with infrequent measurements, we progressively increase sparsity by randomly removing observations while preserving demographic information and labels.

Compared to subsampling, sparsification has a smaller effect on median performance. Across both datasets, AUROC decreases gradually as sparsity increases, but remains substantially higher than in equivalent subsampling conditions.

Physionet-2012: Figure 6 shows the effect of increased sparsity on Physionet-2012. Median AUROC at 10% observation retention (keeping only 10% of measurements per variable) is approximately 0.79–0.81, compared to 0.73–0.76 at 10% cohort size.

MIMIC-III: A similar pattern is observed for MIMIC-III (Fig. 7). Performance at 10% sparsification remains around 0.83–0.85, only slightly below full-data performance. These results suggest that temporal models can compensate for missing observations to a meaningful degree, likely because the remaining measurements still carry informative temporal structure.

However, sparsification introduces increased variance across seeds. At 1–2% observation retention, several models show considerable run-to-run variability (AUROC range > 0.05), indicating that performance becomes unreliable at extreme sparsity even when the median is maintained. GRU-D exhibits a distinct breakdown under extreme sparsification: when only

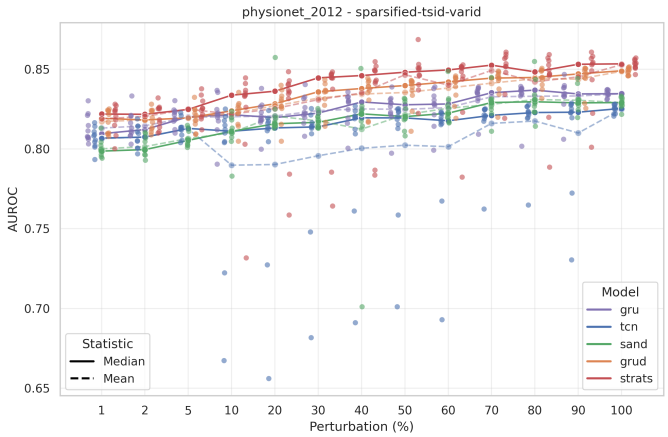

Fig. 6. Model performance (AUROC) under increasing sparsity on Physionet-2012 with varying observation retention levels.

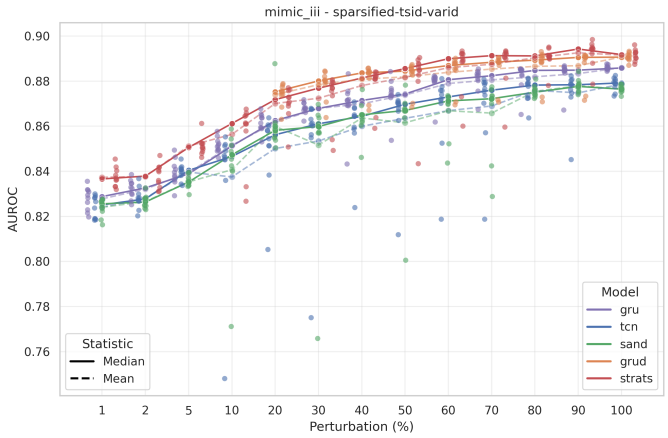

Fig. 7. Model performance (AUROC) under increasing sparsity on MIMIC-III with varying observation retention levels.

a single measurement per variable remains, the model cannot compute meaningful temporal decay values (as Δt requires at least two observations), leading to degraded and unstable outputs. This failure mode is specific to GRU-D's architectural assumptions and does not affect other models.

These results suggest that sparsification primarily degrades stability rather than median performance, with meaningful performance maintained down to approximately 10–20% observation retention. Below this level, instability increases sharply, and GRU-D becomes unreliable.

C. *Effect of Class Imbalance*

To analyze the impact of class imbalance, the length-of-stay (LOS) prediction task is converted into a binary classification problem using varying thresholds.

Class imbalance has a strong negative impact on model performance across both datasets. As the positive class becomes rarer, AUPRC drops substantially and approaches the random baseline (equal to the positive class prevalence) at extreme imbalance levels.

Physionet-2012: Figure 8 shows the impact of class imbalance. Mean AUPRC at 30% positive class rate is approximately 0.55–0.60, declining to 0.15–0.25 at 10% and approaching the random baseline (0.05–0.10) at 1–2%. AUPRC gain over random also diminishes sharply below 20%, indicating that all models struggle to learn meaningful decision boundaries when positive cases are rare.

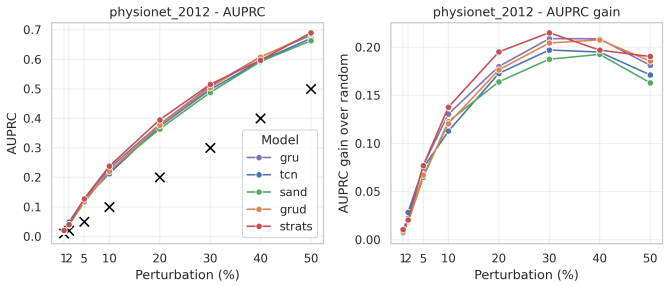

Fig. 8. Model performance (AUPRC) under varying class imbalance on Physionet-2012 across different positive class ratios.

MIMIC-III: A similar trend is observed for MIMIC-III (Fig. 9), where performance decreases with increasing imbalance and approaches the random baseline in extreme cases, indicating a bias toward the majority class.

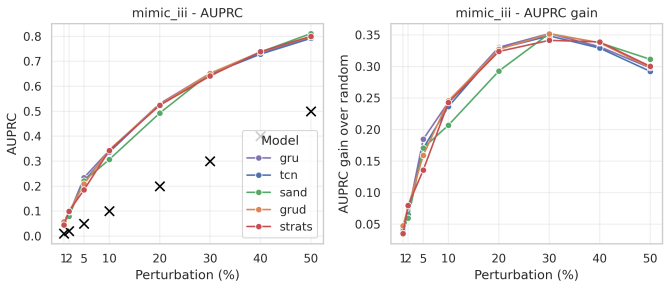

Fig. 9. Model performance (AUPRC) under varying class imbalance on MIMIC-III across different positive class ratios.

Notably, no single architecture exhibits a consistent advantage under severe imbalance. All five models degrade at approximately the same rate, suggesting that class imbalance is a data-level challenge not mitigated by architectural choices within this model family. Cross-validation results (Section IV.D) confirm these trends are stable across data splits.

These results show that class imbalance below 20% positive rate leads to substantial and consistent performance degradation across all evaluated architectures, with no model demonstrating robust behavior under extreme imbalance.

D. *Cross-Validation Analysis on Imbalanced Data*

To further evaluate robustness, we perform 3-fold cross-validation on the imbalanced dataset setting.

Results remain consistent across folds, confirming that observed trends are stable and not dependent on a specific data split. However, overall performance remains limited at low positive class ratios.

At extreme imbalance levels (e.g., $< 10\%$), all models struggle to learn meaningful decision boundaries and tend to favor the majority class.

Physionet-2012: Figure 10 illustrates consistent performance trends across folds.

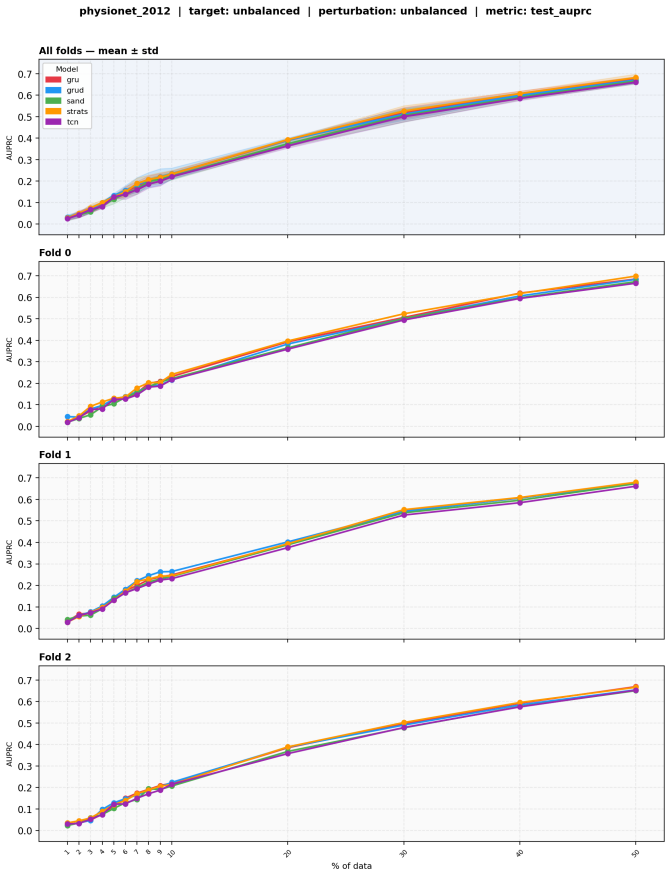

Fig. 10. Cross-validation performance (AUPRC) across three folds on the imbalanced Physionet-2012 dataset.

E. Combined Insights

To evaluate the combined effect of multiple perturbations, we construct a multiperturbed version of the Physionet-2012 dataset that matches the characteristics of a non-ICU cohort (Neutropenic Fever).

Performance drops significantly when the dataset is modified to reflect reduced sample size, increased sparsity, and severe class imbalance.

These findings indicate that performance degradation in non-ICU settings is primarily driven by data characteristics rather than task difficulty.

F. Computational Cost (Runtime Analysis)

We also analyze computational cost by measuring runtime across perturbation settings.

Physionet-2012: Figure 12 shows that GRU-D has significantly higher runtime compared to other models, with cost increasing as dataset size grows.

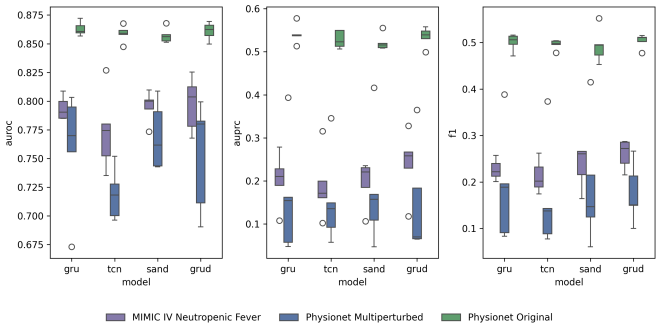

Fig. 11. Comparison of model performance across the original Physionet-2012 dataset, the multiperturbed dataset, and the Neutropenic Fever cohort.

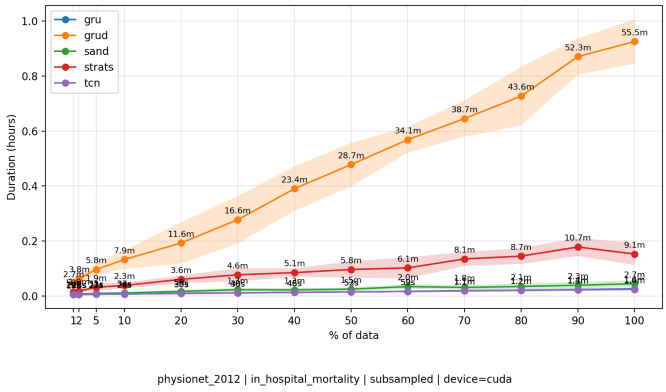

Fig. 12. Runtime comparison of models under subsampling on Physionet-2012 across different dataset sizes.

MIMIC-III: A similar pattern is observed for MIMIC-III (Fig. 13), where GRU-D remains the most computationally expensive model.

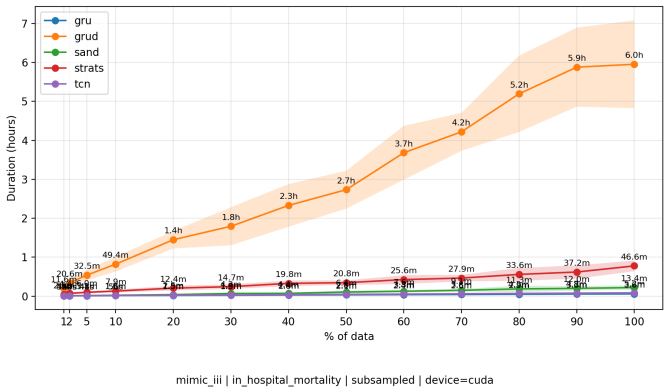

Fig. 13. Runtime comparison of models under subsampling on MIMIC-III across different dataset sizes.

Across all perturbations, runtime differences are primarily driven by model architecture rather than the perturbation itself, with GRU-D consistently showing the highest computational cost.

V. Discussion

A. Summary of Findings

The results of this study demonstrate that temporal deep learning models are highly sensitive to changes in data characteristics commonly observed in non-ICU settings. In particular, model performance declines significantly as the cohort size decreases, with a noticeable drop occurring when the dataset size falls below approximately 2,000 samples.

Similarly, class imbalance has a strong negative impact on performance. As the proportion of the positive class decreases, model performance deteriorates substantially, especially when the positive class falls below 20%. This indicates that models struggle to learn meaningful decision boundaries in highly imbalanced settings.

In contrast, sparsification has a comparatively smaller effect on overall model performance. Although performance decreases slightly with increasing sparsity, the impact is less severe than that of reduced cohort size or class imbalance. However, sparsification introduces instability in some cases, leading to outlier performances.

The differential degradation across architectures reflects the complexity of their inductive biases. STraTS and GRU-D rely on fine-grained temporal structure — through triplet attention and learnable decay terms — which requires sufficient data to learn effectively, making these mechanisms a liability rather than an advantage in low-data regimes. The uniform degradation under class imbalance across all models confirms that this is a data-level challenge that architectural complexity alone cannot resolve.

B. Limitations

This study has several limitations. First, the number of random seeds used in the experiments is relatively low due to computational constraints, which may limit the robustness of the reported results.

Second, cross-validation was applied only to a limited set of experiments, which may affect the generalizability of the findings. A more extensive evaluation across multiple folds and configurations could provide more reliable insights.

Third, the hyperparameters used for model training were adopted from prior work rather than being optimized for each experimental setting. Due to computational limitations, no dedicated hyperparameter tuning was performed, which may have affected model performance.

Finally, certain models, such as GRU-D, rely on the availability of sufficient temporal observations to compute time gaps (Δt). Under extreme sparsification, these models may fail to operate effectively, which limits their applicability in highly sparse data scenarios.

C. Future Work

Future work can extend this study in several directions. First, subsampling can be applied not only to patient IDs but also to variables, in order to better understand the impact of feature availability on model performance.

Second, a systematic combination of perturbations (e.g., jointly applying sparsification, subsampling, and imbalance) could provide deeper insights into how multiple real-world data challenges interact.

Third, performing dedicated hyperparameter tuning for each model and dataset could improve performance and provide a fairer comparison across models.

Finally, the impact of class imbalance can be further addressed using techniques such as random oversampling, undersampling, and weighted loss functions, which may improve model performance in highly imbalanced scenarios.

VI. Conclusion

This study shows that the performance of temporal deep learning models in non-ICU settings is primarily limited by data characteristics rather than model architecture. In particular, reduced dataset size and class imbalance lead to substantial performance degradation, while sparsity mainly introduces instability.

These findings suggest that improving data availability and addressing imbalance should be prioritized over increasing model complexity. Future work should explore techniques such as transfer learning and cost-sensitive training to improve robustness in sparse and imbalanced settings.

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